MPhil Thesis: Introduction

Mohammad-Shah Karim

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1 Introduction

Pharmaceutical innovation is central towards long-run improvements in health and welfare. New drugs can improve quantity of life while changing the treatment possibilities which are available for physicians to administer on patients. At the same time, pharmaceutical innovation is unusually sensitive to policy. Development of a new drug requires a large upfront investment with a long development horizon including patent applications and clinical trials, with only a high probability of failure. Since the returns to a successful drug accrue over multiple years and depend on both prices and the volumes that prevail within the market, the incentives to undertake the investment to develop a drug are shaped not only by opportunity, but also by the economic and institutional environment present which determine the expected future profits once the drug reaches the market.

This makes the pharmaceutical market within the United States of America an ideal setting for studying how public policy shapes scientific opportunity. Federal and state governments are now responsible for roughly half of all prescription drug spending (INSERT SOURCE HERE), both as direct purchasers through programmes such as Medicaid and Medicare, and also indirectly through the tax preferences and regulatory rules that govern private insurance. Medicaid is especially important in this respect - it is a large public buyer, but it does not operate as a standard private insurer. Manufacturers that wish to have Medicaid coverage must participate in the Medicaid Drug rebate Program, and states can use Preferred Drug Lists (PDLs) in order to negotiate additional rebates in exchange for preferred formulary placement. Medicaid demand therefore passes through a public procurement system rather than a conventional insurance market, and its institutional features are described in detail in Section 1.1.

This thesis examines the consequences of one of such reforms for pharmaceutical innovation - the 2014 Medicaid expansion under the the Patient Protection and Affordable Care Act (ACA). The reform produced a well-documented expansion in the size of the Medicaid market, with population coverage increasing by close to twenty million enrollees int he years following the policy change (INSERT SOURCE HERE). Under standard market-sized frameworks in the pharmaceutical sector this should increase innovation incentives for drug-classes more exposed to Medicaid demand (Finkelstein, 2004; Dubois et al., 2015). If firms expect to sell more units after a successful round of innovation, the expected returns to R&D rise. However, the additional demand entered the market through Medicaid's procurement institutions rather than through standard private channels, so an expansion of Medicaid was not simply a positive demand shock for pharmaceutical firms.

The Medicaid setting therefore complicates the prediction under standard pharmaceutical market-sized frameworks. The ACA did not only expand the number of beneficiaries under Medicaid, but it also raised the statutory rebate percentages and extended rebate obligations to Medicaid managed care. Therefore, the reform increased the size of the public market while also increasing the share of revenue back to the state. Additionally, as Medicaid demand grows, preferred placement on a state PDL becomes more valuable. States can use this to negotiate deeper rebates from manufacturers. The same policy may therefore generate two opposing effects. On the one hand, Medicaid expansion raises demand. On the other hand, it may increase pricing pressure and reduce the surplus that firms can capture from that demand.

This thesis therefore raises the question of whether the ACA-induced Medicaid expansion raised pharmaceutical innovation, and whether the response dependent on a firm's exposure to the Medicaid procurement institutions. Put simply, was the expected positive relationship between market size and innovation weakened when the additional demand flows through a state-wide monopsony. Many current drug-pricing reforms, including recent Medicare negotiation policies, rely on the same basic trade-off: lowering public spending and consumer prices may increase static welfare, but it may also affect the expected returns to future innovation, raising the relevance of this question

1.1 Medicaid and the Affordable Care Act

Medicaid is a federal-state public health insurance programme for low-income individuals and other eligible groups in the United States. Prior to 2014 eligibility was largely restricted to specific categories, including low-income children, pregnant women, disabled individuals, and some elderly individuals, as opposed to low-income adults in general. While prescription drug benefits were formally non-mandatory, all state Medicaid programmes provided them. Additionally, outpatient prescriptions are reimbursed through fee-for-service and managed care arrangements.

The ACA substantially expanded this market. From 2014, states were allowed to extend Medicaid eligibility to adults with incomes up to 138% of the federal poverty line. Initially, the expansion was intended to be mandatory, but the Supreme Court's decision in *NFIB v. Sebelius* made this optional for states, hence states adopted the expansion at different times. This threshold change generated a large increase in Medicaid enrolment, and therefore in the number of prescriptions likely to be reimbursed through Medicaid.

Importantly, there are two key institutional features of Medicaid that are noteworthy. The first is the Medicaid Drug Rebate Program (MDRP). Manufacturers that want their drugs to be reimbursed by Medicaid must enter into a national rebate agreement and pay rebates to states for each unit dispensed to Medicaid. Specifically, for brand-name drugs, the basic rebate applied is based on what is greater between a fixed percentage of the average manufacturer price (AMP), or the difference between AMP and the manufacturer's lowest commercial price, known as the "best price". Consequently, discounts offered elsewhere in the market can affect the rebates owed to Medicaid (Duggan and Scott Morton, 2006). Medicaid therefore receives a legally protected discount thereby lowering the value of Medicaid prescriptions to manufacturers.

The ACA also changed the rebate environment directly. The statutory rebate percentage under the MDRP was increased from 15.1% to 23.1% of AMP for most brand-name drugs, and from 11% to 13% for generics. Firms were now obligated to extend rebates to drugs dispensed through Medicaid managed care organisations, while previously being exempt.

The second institutional feature is the Preferred Drug List (PDL). States can negotiate supplemental rebates from manufacturers in exchange for preferred placement on their formulary. Preferred drugs are prescribed to patients without prior authorisation, while non-preferred drugs require additional approval before the prescription is filled. Prior authorisation creates an administrative cost for physicians, therefore preferred placement can affect the prescribing behaviour of a physician and the Medicaid market shares within a therapeutic class as a result. Preferred placement therefore gives states a source of negotiating leverage. Since preferred status affects access conditions and expected utilisation within a therapeutic class, manufacturers may offer supplemental rebates above the statutory MDRP minimum in order to obtain or retain preferred placement. Additionally, some states also participate in multi-state purchasing alliances, which aggregate Medicaid demand across states and may further strengthen their bargaining position.

Selection into the PDL takes on the form of a two-stage selection mechanism. In the first stage, the state's Pharmacy and Therapeutics committee assesses whether a drug is clinically appropriate for preferred status. This involves the use of clinical evidence from peer-reviewed clinical literature and oncology reports to understand the safety, efficacy and effectiveness of the drug relative to to other drugs with the same role. In the second stage, once drugs pass the first stage and are concluded to be therapeutically comparable, states consider the net cost and supplemental rebates when deciding which drugs receive preferred placement. This setting matters as it means that Medicaid procurement does not only depend on clinical value, but also on firms' willingness to offer price concessions. For this reason, an expansion of Medicaid demand may also increase the value of preferred placement and intensify competition over rebates.

Importantly, preferred placement on the PDL is not necessarily limited to a single drug for a given drug role. States may allow for multiple preferred drugs within the same role in order to allow for clinically appropriate choices among patients, such as int the case of route of administration, prior treatment response or clinical needs such as allergies. Furthermore, drugs with non-preferred status are not necessarily from Medicaid coverage, they would instead require authorisation from Medicaid once a physician requests prescribing it.

Thus, the ACA created a demand shock, but it did so inside a public procurement system with rebates and formulary bargaining. For high-Medicaid therapeutic classes, the reform therefore increased the number of prescriptions likely to be paid for by Medicaid, while also increasing the importance of the rebate and PDL mechanisms that determine the net revenue firms receive from those prescriptions.

This thesis does not utilise the staggered timing of state adoption as the main source of variation. Although that variation is useful for studying outcomes such as insurance coverage, healthcare use, or labour supply, it is less suited towards studying pharmaceutical innovation. Pharmaceutical innovation is a national investment decision, and firms cannot develop a drug for sale in one expansion state alone. Instead, they consider expected sales across the United States. Therefore, this thesis treats the start of the ACA Medicaid expansion in 2014 as the relevant policy shock.

1.2 Medicaid Expansion and Pharmaceutical Innovation

The institutional features described change how the standard market-size prediction should be applied to Medicaid expansion. Under the standard framework, an increase in expected demand raises the expected return to successful innovation, and as a result, firms should possess stronger incentives to invest in therapeutic classes where the ACA increased expected prescriptions. This is the market-size channel documented in the pharmaceutical innovation literature (Finkelstein, 2004; Dubois et al., 2015).

In this setting, however, the relevant quantity is not Medicaid demand but the surplus firms expect to gain from that demand. Medicaid prescriptions are subject to statutory rebates, and gaining preferred placement on a state PDL can require additional price concessions through supplemental rebates. Therefore, the expansion raised expected prescription volume while also increasing the importance attached to the institutions determining what proportion of said revenue that firms can retain. The net effect on innovation is therefore ambiguous: it may be positive if the demand effect dominates, but negative if procurement institutions reduce expected returns sufficiently.

This thesis studies that ambiguity empirically through three research questions. First, this paper examines whether therapeutic classes with greater pre-expansion exposure to Medicaid experienced differential innovation responses following 2014, using patents and FDA approvals as outcome measures. This tests whether the standard market-size prediction holds when the demand shock operates through Medicaid. Contrary to standard predictions, the results suggest that therapeutic areas more exposure to Medicaid experienced declining rather than increasing patenting after expansion.

Second, Medicaid demand exposure is separated from procurement exposure by incorporating PDL status into the empirical analysis. This allows for testing on whether innovation responses differed between firms and classes that are more exposed to Medicaid demand and those that are more exposed to Medicaid's formulary mechanism. Thirdly, this paper examines whether the effect of Medicaid exposure varies across therapeutic areas, allowing the analysis to assess for heterogeneity across broader therapeutic classes.

To interpret these empirical results, counterfactual analysis is conducted for the PDL, along with the construction of a theoretical model for firms' innovation responses. The counterfactual analysis exploits the two-stage structure of PDL selection in order to distinguish between clinical and pricing margins of preferred status. The theoretical model then formalises the mechanism by which the Medicaid expansion raised gross demand while also raising the procurement wedge faced by firms. Together, these exercises help explain why an expansion in Medicaid demand may have not uniformly led to an increase in pharmaceutical innovation.

The closest related paper to this work is Garthwaite et al. (2021), which studies the effect of the ACA-induced Medicaid expansion on clinical trial activity and finds little evidence of an aggregate innovation response. This thesis builds on this work in three ways. First, it uses two alternative measures of innovation: patents and FDA approvals. Patenting activity captures earlier stage innovation outcomes, while FDA approvals capture the final stage regulatory output. This allows the analysis to examine whether the Medicaid expansion affected different points of the innovation pipeline. Secondly, it extends the analysis through to 2025, allowing for a longer post-expansion window, as opposed to sticking with 2016. Pharmaceutical development timelines are long, and a policy shock in 2014 may take several years to appear in patenting or approval activity. Third, it moves from therapeutic-class-level analysis to a firm-by-class analysis, linking innovation outcomes to Medicaid demand exposure and PDL exposure. This makes it possible to study whether the effect of the Medicaid expansion varies with a firm's exposure to the Medicaid formulary process. The empirical results then inform the theoretical model developed later in the thesis, which studies how the Medicaid expansion affected firms' innovation incentives when PDL procurement changes the surplus firms expect to capture.

A substantial portion of this thesis is constituted by the construction of the dataset - a key contribution of this paper. The empirical analysis requires linking innovation outcomes, Medicaid demand exposure, and PDL exposure across datasets that do not share a common identifier. Firms are linked using National Drug Code labeler information, FDA sponsor names, patent assignees, and text-based name matching. Drug names and active ingredients are harmonised to therapeutic classes using RxClass and the World Health Organisation's ATC hierarchy. The resulting panel makes it possible to study Medicaid procurement exposure at the firm-by-class level over the period 2010 to 2025.

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